serviceDiscovery

Service Description

Service ID: *"serviceDiscovery"*

Abstract
This document describes the `serviceDiscovery` service, version 1.0.0, produced by the ServiceRegistry core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceDiscovery` service, which provides registration and lookup of service instances: which provider offers which service definition, in which version, through which interfaces and with which security policy. It is produced by the ServiceRegistry system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Successor of the Arrowhead 4.x ServiceRegistry `register`, `unregister` and `query` operations. In 5.x a service instance is identified by `<System>|<serviceDefinition>|<version>`, and interfaces are described by interface templates instead of fixed interface names.

2.2 How This Service Is Meant to Be Used

A provider calls `register` for each service it offers (creating or replacing its instance), and `revoke` when it stops offering it. Consumers normally do not call `lookup` directly: they ask an orchestrator, which looks up the candidates. With `service.discovery.policy=restricted` (default) a direct `lookup` of other than the core services is reserved to sysops and the systems in `service.discovery.direct.access`.

2.3 Important Delimitations and Dependencies

A system can only register services under its own name. In this workspace the service definitions of the core and support systems are reserved for their owning systems (`service.definition.reserved`, finding J9); other registrations of them are refused with 403. Interfaces must conform to a known interface template unless `service.discovery.interface.policy` is `extendable` or `open`. Registration says nothing about authorization: whether a consumer may use the instance is decided by ConsumerAuthorization.

3 Service Interface

This section describes the interfaces to the `serviceDiscovery` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `serviceDiscovery` service and its operations.

The following interface operations are available.

3.1 operation register (ServiceInstanceCreateRequest) : ServiceInstanceResponse

Registers a service instance of the requesting system. An existing instance with the same identifier is replaced. Unknown service definitions are created; interfaces are validated against their templates.

3.2 operation lookup (ServiceInstanceLookupRequest) : ServiceInstanceListResponse

Returns the service instances matching all given filters. Under the restricted discovery policy, a requester without direct access only receives instances whose service has the `unrestrictedDiscovery` metadata (the core services). With `verbose=true` the provider and device details are included.

Optional parameter: `verbose`.

3.3 operation revoke (Identifier)

Removes a service instance of the requesting system. Returns 200 if removed, 204 if it did not exist.

4 Information Model

Here, all data objects that can be part of the `serviceDiscovery` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.12, which are used to represent things like hashes and identifiers.

4.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

4.2 struct DeviceResponse

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.3 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.4 struct ServiceDefinitionResponse

A service definition.

Field	Type	Description
name	Name	Service definition name.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.5 struct ServiceInstanceCreateRequest

Registers a service instance of the requesting system.

Field	Type	Description
serviceDefinitionName	Name	Service definition name, camelCase (e.g. `temperatureReading`).
version	Version	Version of the service instance (default `1.0.0`).
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Request>	Service interfaces offered by the instance.

4.6 struct ServiceInstanceInterfaceRequest

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface (`NONE`, `CERT_AUTH`, token policies).
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.7 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.8 struct ServiceInstanceListResponse

List of service instances.

Field	Type	Description
entries	List<ServiceInstanceResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.9 struct ServiceInstanceLookupRequest

Filter for service instances; all fields optional, combined with AND.

Field	Type	Description
instanceIds	List<Identifier>	Instance identifiers to match.
providerNames	List<Name>	Provider system names to match.
serviceDefinitionNames	List<Name>	Service definition names.
versions	List<Version>	Service versions to match.
alivesAt	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
metadataRequirements List	List<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
addressTypes	List<String>	Address types the interfaces must offer (`HOSTNAME`, `IPV4`, `IPV6`, `MAC`).
interfaceTemplateNames	List<String>	Accepted interface templates, e.g. `generic_https`.
interfaceProperty RequirementsList	List<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
policies	List<String>	Interface security policies to match, e.g. `CERT_AUTH`.

4.10 struct ServiceInstanceResponse

A registered service instance.

Field	Type	Description
instanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
provider	SystemResponse	The provider system.
serviceDefinition	ServiceDefinitionResponse	The service definition.
version	Version	Version of the service instance.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.11 struct SystemResponse

A registered system.

Field	Type	Description
name	Name	System name.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<Address>	Network addresses with their types.
device	DeviceResponse	The device hosting the system, if registered.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.12 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceregistry/documentation/v5.3.0/ServiceRegistry-v5.3.0-SysD.pdf.

[2] ——, “serviceDiscovery HTTP/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-serviceDiscovery-HTTP-IDD.pdf.

[3] ——, “serviceDiscovery MQTT/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-serviceDiscovery-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	